

DLSP: A contrastive learning framework for disentangling long- and short-term preferences in next POI recommendation

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ABSTRACT

Next point-of-interest (POI) recommendation aims to suggest locations that align with users' preferences. User behavior is typically influenced by both long-term preferences, which capture periodic behavioral patterns, and short-term preferences, which reflect immediate contextual needs. However, existing methods often model these two types of preferences in a unified manner, leading to weakened individual representations and negatively impacting recommendation performance. To address this issue, we propose the Disentangle Long- and Short-term Preferences (DLSP) model for next POI recommendation. DLSP utilizes separate encoders to enhance the representation of long-term and short-term preferences, enabling more expressive semantic modeling for each. A dynamic fusion mechanism is introduced to effectively integrate the two types of preferences. Additionally, a user enhancement module is designed to emphasize the influence of the latest contextual dynamics on user decisions. Experimental results on three real-world datasets demonstrate that our DLSP model consistently outperforms state-of-the-art techniques, achieving average improvements of 11.2 % in HR@5 and 15.1 % in NDCG@5. Our source code is publicly available at <https://github.com/belle-0/DLSP>.

1. Introduction

Location-based social networks (LBSNs) like Foursquare and Yelp are commonly used by the public, where users on LBSNs share their point-of-interest (POI) footprints with their friends. The vast amount of user check-in data generated on these platforms has spurred extensive research into POI recommendation systems. Beyond enhancing user satisfaction, accurate POI recommendation holds significant real-world value: it can boost local economies by guiding foot traffic, support personalized travel services, and inform intelligent urban planning [1–4]. Therefore, optimizing these systems brings not only technical progress but also tangible social and economic benefits.

The core challenge in this domain lies in modeling the complex nature of user preferences, which are a mixture of two distinct dimensions: stable **long-term preferences** and dynamic, context-driven **short-term preferences**. Long-term preferences are often reflective of a user's stable behavioral patterns, such as daily routines, enduring habits, and consistent interests. These preferences provide a broad view of the user's overall inclinations over an extended period. Conversely, short-term preferences capture dynamic and context-specific needs, representing immediate interests or situational priorities. This distinction

enables the system not only to consider the user's overall behavioral characteristics but also to respond to their specific needs when making recommendations.

For instance, the behavioral trajectory of a user exemplifies the synergy between long-term and short-term preferences (Fig. 1). His long-term preferences manifest in consistent routines—commuting to the office on weekdays, attending fitness classes and frequenting a nearby bookstore—reflecting his stable lifestyle choices and interests. Meanwhile, his short-term preferences are shaped by contextual factors such as weather, spontaneous cravings, or event-specific motivations. On weekends, for example, he might explore a park or attend a live event depending on real-time conditions. This fluid interaction between enduring habits and momentary needs underscores the necessity of jointly modeling both preference dimensions to support accurate, timely and personalized POI recommendation.

A major challenge in existing models is the lack of explicit mechanisms to distinguish and integrate long-term and short-term preferences effectively. Without clear separation, these preferences often interfere with each other during the learning process, reducing the accuracy and reliability of recommendation outcomes. While several works attempt

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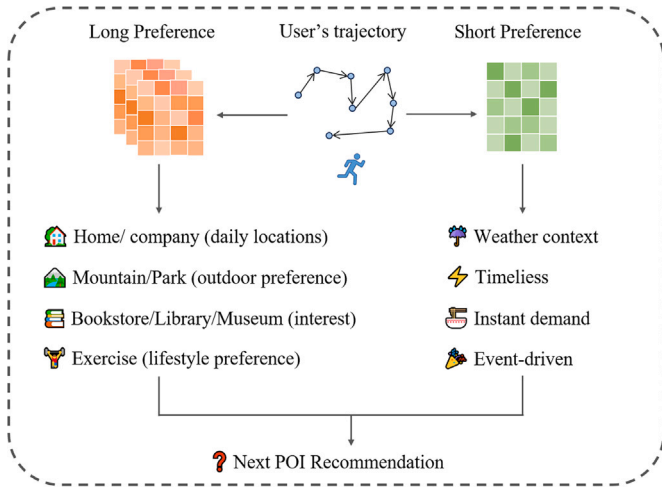


Fig. 1. The user's behavioral trajectory reflects the interplay between enduring routines and transient contextual influences.

to capture both preference types, they often fail to achieve effective disentanglement. For instance, Zhang et al. [5] integrate Mogrifier LSTM and multi-head attention to capture spatio-temporal dependencies. Wu et al. [6] employ a dual-channel framework to model long- and short-term preferences separately. Recent works leverage contrastive learning to explicitly address this issue. However, they often suffer from suboptimal architectural designs. For instance, Duan et al. [7] and Liu et al. [8] use a shared encoder together with contrastive objectives to emphasize commonalities between the two sequences. While this shared-encoder design facilitates the learning of shared patterns, it inherently limits the model's ability to capture the distinct dynamics of each preference type. Moreover, preference integration remains challenging: existing fusion strategies are often either naive (simple concatenation, e.g., [9,10]) or computationally expensive and hard to interpret (MLPs or attention-based modules, e.g., [11,12]). Therefore, a significant challenge remains: to design a fusion mechanism that is effective and capable of adaptively integrating the two preference signals in a lightweight and principled manner.

To tackle the issue of feature interaction between long- and short-term preferences, we develop a contrastive learning-based framework that explicitly decouples the two signals through separate, specialized encoders. Specifically, we employ a Transformer to capture the global patterns in a user's long-term history and a GRU to model the strictly sequential nature of their short-term daily trajectory. A contrastive loss function is then introduced to enforce this separation, promoting representational independence while allowing for their precise and independent representation. Building upon this robust decoupling, we design a dynamic fusion mechanism that adaptively adjusts the relative weights of the long- and short-term preferences based on user behavior. For actions predominantly influenced by long-term preferences, the model assigns them higher weights, and conversely, shifts focus when short-term preferences take precedence. Finally, the fused representations are further refined and contextualized with personalized user embeddings via a Temporal Convolutional Network (TCN) [13] to generate accurate, user-aligned recommendations. Through this approach, the model achieves a balanced integration of long-term and short-term preferences across various contexts, thereby improving both the precision and adaptability of the recommendations.

In summary, the main contributions of this paper are:

- We propose a dual-encoder architecture that utilizes a Transformer to capture long-term preferences and a GRU to model the short-term preferences. This specialized design effectively mitigates

representational interference and enables explicit disentanglement of long- and short-term preferences.

- We develop a contrastive learning objective applied to the representations produced by the dual encoders. This objective explicitly enforces distinction and complementarity between the two preference embeddings, addressing the limitations of prior methods that rely on implicit or entangled separation.
- We introduce a dynamic fusion gate that balances the contributions of the disentangled preferences. This mechanism provides a computationally efficient strategy, outperforming naive concatenation and avoiding the high cost and low interpretability of complex MLP- or attention-based fusion approaches.
- We perform extensive experiments and comparative analyses with multiple baseline models. The results demonstrate the superior performance of our approach and provide deeper insights into its effectiveness.

2. Related work

2.1. General POI recommendation

Early studies mainly focused on temporal dependencies using sequential models, such as ST-RNN [14] and DeepMove [15], which capture sequential patterns of user check-ins. Some early spatial models considered only geographical proximity or distance between POIs, without fully integrating temporal or contextual factors. Wang et al. [16] addressed the non-symmetric and highly variable geographical influence between POIs by introducing POI-specific geo-influence and geo-susceptibility factors. Later, researchers incorporated both spatial and contextual information, leading to spatio-temporal frameworks. For instance, Zhao et al. [17] extended LSTMs by introducing time and distance gates that explicitly encode the spatio-temporal intervals between consecutive check-ins, while He et al. [18] employed a specialized attention mechanism to capture irregular temporal and spatial intervals within user trajectories. Cheng et al. [19] utilized a Location-aware Topic Model to semantically connect music content with different types of venues, ensuring recommended songs match the current location's context. More recent attention-based models further improve the ability to capture complex spatio-temporal dependencies and long-term user preferences [20–22].

2.2. Long- and short-term preference modeling

Recent studies have focused on modeling long- and short-term preferences independently to improve recommendation accuracy. For example, Ying et al. [23] introduced a hierarchical attention mechanism to capture both types of preferences effectively. Sun et al. [24] utilized an LSTM-based model with a non-local network for long-term preferences while extending RNNs for short-term sequences. Similarly, Liu et al. [25] leveraged graph neural networks to embed POIs, combining both preference types for a more comprehensive representation. Wang et al. [26] introduced a multi-level attention mechanism to model the dynamic representation of check-ins and category transitions, while Wang et al. [12] proposed a Hierarchical LSTM framework that jointly models POI and semantic trajectories to enhance long- and short-term preference learning. Zuo et al. [27] integrated separate graphs to capture POI features from both global (inter-POI) and user (historical) perspectives. Jiang et al. [28] adaptively fused long- and short-term preferences by exploiting spatio-temporal context. Zhao et al. [29] introduced mid-term, short-term and long-term preferences, using a gated fusion mechanism to integrate the representations from these three temporal dimensions. Li et al. [9] integrated multi-dimensional information using separate modules for long- and short-term preferences. While these models have made strides in capturing users' long-term and short-term preferences, they often fall short in effectively distinguishing and accurately delineating the two, limiting their ability to leverage the unique characteristics of each preference type.

2.3. Contrastive learning in recommendation

Self-supervised learning (SSL) is a powerful paradigm that extracts transferable knowledge from unlabeled data, achieving great success in computer vision [30,31], natural language processing [32,33], and graph learning [34]. These successes have motivated researchers to explore its potential in recommender systems. The application of SSL in recommender systems can be broadly categorized into three main approaches: generative, contrastive, and predictive. Since generative methods are resource-intensive and predictive ones lack generalization [35], contrastive learning has become the most popular approach, favored for its simplicity and effectiveness in learning informative representations without manual labels.

A representative work, CL4SRec [36], applied data augmentation techniques such as cropping, masking, and reordering to generate diverse sequence views. Subsequent studies extend contrastive learning to capture user preferences from multiple perspectives. For example, Duan et al. [7] employed a shared-parameter encoder to separately model long- and short-term sequences. Qin et al. [37] employed meta-optimization for adaptive contrastive training. Yang et al. [38] introduced consistency-aware debiasing to integrate sequential and global contexts. Zhang et al. [39] further improved representation quality through dual contrastive objectives across feature and batch spaces. Chen et al. [10] utilized frequency and hierarchy to enhance user preference representations, addressing the loss of high-frequency information caused by the Transformer model acting as a low-pass filter. Beyond sequential modeling, graph contrastive learning (GCL) has emerged as a powerful technique for general graph representation learning. For instance, Sun et al. [40] refined the selection of positive and negative pairs by modeling cross-view information flow, and Peng et al. [41] enhanced generalization by incorporating explicit constraints from downstream tasks. Zhang et al. [42] mitigated the sparsity issue in check-in data by exploiting potential behavioral commonalities between long- and short-term sequences. Building on these foundations, recent studies have introduced GCL into the POI recommendation domain to better capture spatial and temporal dependencies. For example, Qin et al. [43] separated sequential and geographical influences using a dual-graph contrastive framework, while Fu et al. [44] proposed a self-supervised model that fuses short-term and long-term geographic interests. Despite these advancements, the full potential of contrastive learning in next POI recommendation remains underexplored. Current methods often fail to fully leverage disentangled representations of long-term and short-term preferences, resulting in overlapping features and reduced recommendation accuracy.

3. Problem definition

The definitions of the main terms used in our research problem are as follows. Let $L = \{l_1, l_2, \dots, l_M\}$ be a set of POIs; $U = \{u_1, u_2, \dots, u_N\}$ denote a set of users; $C = \{c_1, c_2, \dots, c_P\}$ be a set of categories of POIs, where M , N , and P are respectively the total number of POIs, users and categories. And let $T = \{t_1, t_2, \dots, t_{24}\}$ represent a collection of time slots, with each day divided into 24 slots. Since temporal information is assumed to be given in many location prediction scenarios, users will be in a different state of novelty-seeking at different times, e.g., they may be more inclined to explore during weekends. Thus, let d_0 represent weekdays and d_1 represent weekends. Therefore, a check-in record $r = (u, l, c, g, t)$ signifies that user u visited the POI l at the time t . The POI l is defined by its category c and geocoded by g (longitude and latitude).

To effectively differentiate between a user's long-term and short-term preferences, we follow the method outlined in [45], segmenting each user's trajectory into two parts: a long-term sequence representing consistent behavioral trends and a short-term sequence capturing immediate preferences.

(1) **Trajectory:** We define $Q_i^u = \{r_1, r_2, \dots, r_{|Q|}\}$ to denote all the records of user u on day i . $|Q|$ denotes the total number of locations that the user visits in a single day.

Table 1
Summary of key notations.

Symbol	Description
r	A check-in record
e_i	Embedding of check-in record r_i
Q_i^u	All check-in records of user u on day i
T_i^u	Long-term daily check-in sequences of user u before the i -th day, i.e., $\{Q_1^u, Q_2^u, \dots, Q_{i-1}^u\}$
T_s^u	Daily check-in sequence of user u on the i -th day, i.e., Q_i^u
$E_{Q_i^u}$	The embedding of the check-in sequence of user u on the i -th day
$E_{T_i^u}$	Historical daily embeddings of user u before the i -th day, i.e., $\{E_{Q_1^u}, E_{Q_2^u}, \dots, E_{Q_{i-1}^u}\}$
$E_{T_s^u}$	Embedding of the current day's check-in sequence of user u , i.e., $E_{Q_i^u}$
h_i^u	Representation of the long-term sequence of user u encoded from $E_{T_i^u}$
h_s^u	Representation of the short-term sequence of user u encoded from $E_{T_s^u}$
p_i^u	Average-pooled representation of $E_{T_i^u}$
p_s^u	Average-pooled representation of $E_{T_s^u}$

(2) **Long-term Sequence:** Long-term preferences refer to the user's preferences in the past. Here, we denote the sequence of check-ins before being predicted as a long-term preference sequence, denoted as $T_i^u = \{Q_1^u, Q_2^u, \dots, Q_{i-1}^u\}$.

(3) **Short-term Sequence:** Short-term preferences are used to represent the user's recent preferences. We define the sequence of check-ins on the day being predicted as a short-term preference sequence, denoted as $T_s^u = Q_i^u$.

For each user, our task is to forecast the next POI they will visit, using their past check-in sequence for prediction. Frequently used symbols are summarized in Table 1.

4. Methodology

Fig. 2 presents the architecture of our proposed model, designed to effectively address long-term and short-term user preferences. The framework includes separate encoders for capturing long-term and short-term sequences, a contrastive learning module to disentangle preference representations, and a user enhancement module to refine personalized embeddings. To improve recommendation accuracy, an attention-based fusion mechanism dynamically adjusts the weights of outputs from the two encoders. The fused representations are further combined with the user enhancement module's features to compute the final POI loss, guiding the optimization process and enhancing prediction performance.

4.1. Trajectory modeling

Trajectory Embedding. When constructing accurate and efficient POI recommender systems, the content of the dataset used is very rich. These datasets not only contain the interaction records between users and POIs, but also contain rich contextual information, which has a decisive impact on enhancing the effectiveness of the model. Therefore, we will encode the embedding such that $r = (u, l, c, g, t)$ represents a check-in containing contextual information, including user, POI location, category, geographical coordinates (latitude and longitude), and access time. This ensures that these elements are effectively captured and utilized within the model.

For the access record $Q_i^u = \{r_1, r_2, \dots, r_{|Q|}\}$ of user u on day i , each of these location access records r_i can be embedded as:

$$e_i = \mathbf{u}_i \oplus \mathbf{p}_i \oplus \mathbf{c}_i \oplus \mathbf{t}_i \oplus \mathbf{w}_i \quad (1)$$

where $\oplus$ denotes the concatenation operation, $\mathbf{u} \in \mathbb{R}^{d_u}$ denotes the embedding of the user, $\mathbf{p} \in \mathbb{R}^{d_p}$ denotes the embedding of the location of the POI, $\mathbf{c} \in \mathbb{R}^{d_c}$ denotes the embedding of the category of the POI, $\mathbf{t} \in \mathbb{R}^{d_t}$ denotes the embedding of the specific time of the check-in, $\mathbf{w} \in \mathbb{R}^{d_w}$ represents the embedding indicating whether the check-in time falls on a weekend. The embedding of a trajectory $Q_i^u = \{r_1, r_2, \dots, r_{|Q|}\}$ is denoted

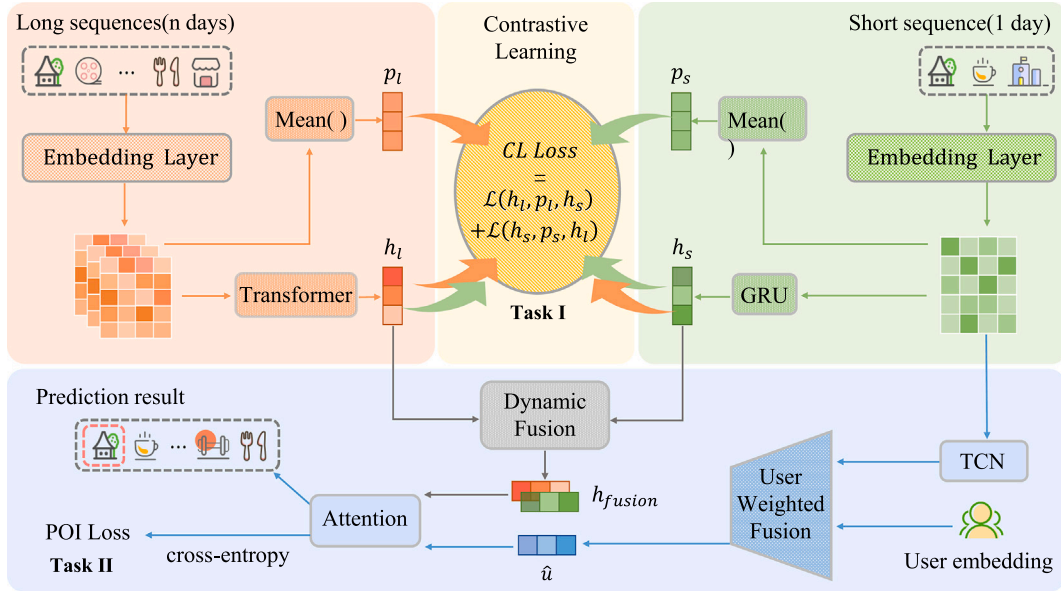


Fig. 2. Overview of our proposed Disentangle Long- and Short-term Preferences (DLSP) model for POI recommendation.

as $E_{Q_i^u} = \{e_1, e_2, \dots, e_{|Q|}\} \in \mathbb{R}^{|Q| \times D}$, where the simple embedding layer is used.

Long Trajectory Modeling. We investigate the use of a trajectory encoder to model users' long-term preferences by analyzing their behavioral patterns over multiple consecutive days. However, two key challenges arise during this modeling process. First, users' check-in behaviors are often influenced by external episodic factors such as holidays, weather changes, or special occasions, which can introduce noise and obscure their true travel or location-based preferences. Second, the inherent complexity and diversity of user behavioral trajectories, combined with individual differences in habits, can undermine the model's robustness and generalizability, making it difficult to effectively capture long-term preferences across varied user groups.

To address these challenges, we propose a masking strategy inspired by the work in [7]. This approach simulates a scenario where interference from external events is mitigated by randomly masking or replacing segments of the user's long-term behavioral sequence. The purpose of this strategy is to filter out noisy behavior caused by episodic events, allowing the model to better capture users' inherent and stable travel preferences. Additionally, this masking technique enhances the model's flexibility in adapting to complex behavior patterns, enabling it to accurately identify long-term preferences despite the variability in user behavior. As a result, the proposed strategy improves the model's robustness and generalization across diverse user groups. For the long trajectory $T_i^u = \{Q_1^u, Q_2^u, \dots, Q_{i-1}^u\}$ of user u , a random mask is performed and then input to the embedding layer to obtain the embedding $E_{T_i^u} = \{E_{Q_1^u}, E_{Q_2^u}, \dots, E_{Q_{i-1}^u}\}$. Next, we feed each day's trajectory into the long trajectory encoder. Here, we use the mechanism of the Transformer as the long encoder.

$$h_{Q_i^u} = \text{Transformer}(E_{Q_i^u}) \quad (2)$$

Transformer operates as follows:

$$h_{Q_i^u} = \text{LayerNorm}(E_{Q_i^u} + \text{Dropout}(\text{FFN}(M_i))) \quad (3)$$

$$M_i = \text{Attention}(E_{Q_i^u} W_q, E_{Q_i^u} W_k, E_{Q_i^u} W_v) \quad (4)$$

where $W_q, W_k, W_v \in \mathbb{R}^{D \times D}$ are trainable parameter matrices respectively representing query, key and value. Thus the long-term trajectory of user

u is represented as:

$$h_l^u = h_{Q_1^u} \oplus h_{Q_2^u} \oplus \dots \oplus h_{Q_{i-1}^u} \quad (5)$$

where $\oplus$ denotes the concatenation operation, which concatenates together the user's daily trajectory encoding, resulting in the long-term trajectory encoding h_l^u for user u .

Short Trajectory Modeling. To effectively distinguish between users' long-term and short-term preferences, we employ the gated recurrent unit (GRU) [46] as the encoder for short-term sequence modeling. GRU's computational efficiency and adaptability make it well-suited for processing dynamic and temporal data. This design captures evolving patterns and temporal dependencies in users' same-day check-in sequences, reflecting immediate travel intentions and rapidly changing demands. Using the cyclic structure and gating mechanism of GRU, the model dynamically updates the user states in real time, enabling the precise extraction of short-term behavioral patterns. These short-term features are seamlessly integrated with long-term preferences modeled by the trajectory encoder, forming a comprehensive representation of user behavior. This integration enhances the system's ability to deliver personalized and context-aware recommendations, effectively addressing both persistent habits and situational needs.

For the short-term sequence $T_s^u = Q_i^u$ for user u , since it is embedded only for one day for the user, we input it directly into the embedding layer to obtain $E_{T_s^u}$, which is then fed into the short-term encoder:

$$h_s^u = \text{GRU}(E_{T_s^u}) \quad (6)$$

Here, GRU is used to reflect the user's current state and changes in demand, thereby obtaining a short-term sequence encoding for the user.

4.2. Self-supervised disentanglement of long- and short-term preferences

When analyzing user behavior patterns, long-term and short-term trajectories reflect different but interconnected aspects of user preferences. Long-term trajectories offer a macro perspective on a user's consistent behavioral tendencies, such as residential habits, lifestyle interests, and frequent destinations, while short-term trajectories capture momentary intentions and contextual decisions. For instance, an outdoor sports enthusiast may frequently visit hiking and climbing locations over time, suggesting stable long-term interests. However, their

short-term trajectory may change due to weather, seasonal activities or one-time events, reflecting the dynamic nature of user behavior.

Rather than assuming that these two types of preferences are strictly oppositional, we treat them as complementary but distinguishable components of user interest. To better capture their nuanced relationship, we introduce a contrastive learning framework that promotes representational differentiation between the long-term and short-term preference encoders. This strategy encourages each encoder to focus on its respective behavior pattern while maintaining potential correlations. To support the contrastive objective, we construct interest agents from users' interaction sequences—proxy representations that summarize historical and recent behavior, respectively. These agents provide lightweight supervision anchors to guide the encoding process. This approach enables the model to remain both structurally disentangled and behaviorally adaptive, enhancing its ability to track shifts in user preferences and improve downstream performance.

By leveraging these interest agents, the training process of the preference encoders is guided and optimized based on actual user interactions. The computation is defined as follows:

$$p_l^u = \text{Mean}(E_{T_l^u}) \quad (7)$$

$$p_s^u = \text{Mean}(E_{T_s^u}) \quad (8)$$

where $\text{Mean}(E_{T^u})$ represents taking the mean over all embeddings in the sequence.

Contrast Strategy. By employing agents as labels, we can use them to supervise the decoupling of long- and short-term preferences. Specifically, we perform contrastive learning between encoder outputs and agents, ensuring that the learned depictions of these two types of preferences align more closely with their respective agents than with opposing agents. Consequently, the contrastive learning task is defined as follows:

$$\text{sim}(h_l^u, p_l^u) > \text{sim}(h_l^u, h_s^u) \quad (9)$$

$$\text{sim}(h_s^u, p_s^u) > \text{sim}(h_s^u, h_l^u) \quad (10)$$

The contrastive loss can be pre-defined as:

$$\mathcal{L}(a, b, c) = -\log(\sigma(a, b)) - \log(1 - \sigma(a, c)) \quad (11)$$

This loss $\mathcal{L}(a, b, c)$ aims to maximize the similarity between positive pairs of the same user while minimizing the similarity between positive and negative samples. The increased similarity of positive pairs should be the key to capturing the intrinsic consistency of user preferences, while distinguishing negative samples should enhance the model's robustness by mitigating noise or irrelevant features. Following these steps, we define the contrastive learning loss function as:

$$\mathcal{L}_{con} = \mathcal{L}(h_l^u, p_l^u, h_s^u) + \mathcal{L}(h_s^u, p_s^u, h_l^u) \quad (12)$$

4.3. Dynamic and user weighted fusion

Dynamic Fusion of Long- and Short-Term Preferences. Effectively combining decoupled long-term and short-term preferences remains a critical challenge for accurate predictions. To tackle this, we propose a dynamic fusion strategy that adjusts the influence of each preference type based on their similarity. This strategy leverages cosine similarity to quantify the alignment between long-term and short-term preference vectors, enabling adaptive weighting to better reflect user behavioral tendencies.

Specifically, we first quantify the similarity between the long-term and short-term preferences using weighted cosine similarity, which captures the directional alignment between the two preference vectors. Once the similarity is calculated, it serves as a weighting coefficient in a

linear combination of these preferences. When the similarity is high, the model places greater emphasis on long-term preferences, as this reflects a stronger alignment with the user's consistent and stable interest patterns. Conversely, when the similarity is low, the model assigns higher weight to short-term preferences, better capturing the user's current, context-driven needs and interest fluctuations. The specific formula of this fusion strategy is as follows:

$$\alpha = \frac{\sum_{i=1}^n w_i \cdot (h_l^u)_i \cdot (h_s^u)_i}{\sqrt{\sum_{i=1}^n w_i \cdot (h_l^u)_i^2} \cdot \sqrt{\sum_{i=1}^n w_i \cdot (h_s^u)_i^2}} \quad (13)$$

$$\beta = 1 - \alpha \quad (14)$$

$$h_{fusion}^u = \alpha h_l^u + \beta h_s^u \quad (15)$$

where α is the weighted cosine similarity between long-term and short-term preferences, and h_{fusion}^u is a fused representation of the user's long-term and short-term preferences.

User Weighted Fusion. In the preceding sections, we effectively extracted users' long-term and short-term preferences through a trajectory encoder, enabling a detailed investigation into the similarities and differences between these two types of preferences with the aid of contrastive learning. Furthermore, we recognize that short sequence data plays a dual role: it not only uncovers latent user preferences that tend to be context-dependent but also serves as a direct and immediate representation of the user's current state, capturing their situational needs and transient interests in real time.

To accurately capture these state characteristics, we learn the user's short sequences using a TCN, which has demonstrated strong advantages in sequence data processing due to its superior parallel processing capability and effective modeling of long-term dependencies. Specifically, TCN achieves this by utilizing causal and dilated convolutions. This design ensures a unidirectional flow of information while maintaining low parameter complexity. Therefore, we apply TCN to learn users' short-term sequences to dig deeper and understand their current state. Through the layer-by-layer processing of TCN, we are able to capture the user behavioral patterns over time in the short-term sequences and then extract the key features reflecting the user's current state. Subsequently, to construct a richer and more comprehensive user representation, we fuse the short-term sequence features obtained from TCN encoder with user feature embeddings:

$$\hat{u} = \gamma \text{Embedding}(u) + (1 - \gamma) \text{TCN}(h_s^u) \quad (16)$$

where γ takes different values for different datasets. Finally, the augmented user representation is concatenated together with a comprehensive feature representation that fuses both long-term and short-term user preferences, creating a unified input for downstream tasks.

4.4. Loss function

After obtaining the enhanced user representation, we employ an attention mechanism to integrate it with the vector that combines both long-term and short-term features. Then, a two-layer MLP is used to predict the final result. The prediction result $\bar{y}_i$ is calculated by:

$$\bar{y}_i = \text{MLP}(\text{Attention}(h_{fusion}^u W_q, \hat{u} W_k, \hat{u} W_v)) \quad (17)$$

During the model training process, cross-entropy is used as the loss function for predicting POIs:

$$\mathcal{L}_{poi} = - \sum_{i \in N} \log(\bar{y}_i) \quad (18)$$

where $\bar{y}_i$ denotes the predicted POIs to be visited at the next time, and N refers to the count of POIs. The cross-entropy loss function, commonly used in classification tasks, measures the difference between

Table 2
Dataset details.

Dataset	#POI	#User	#Check-in	#Density
PHO	7247	2946	47,980	0.22%
SIN	33,712	8648	355,337	0.12%
NYC	56,252	16,387	511,431	0.06%

predicted and true labels. By minimizing this difference, the model optimizes its parameters, enhancing prediction accuracy. Well-suited for multi-classification tasks, cross-entropy loss is ideal for our POI prediction scenario, where the goal is to identify the most probable next POI from multiple candidates. Nevertheless, relying on cross-entropy loss exclusively could result in inadequate model generalization, particularly when addressing the unique requirements of diverse users and evolving user preferences. To address this challenge, we propose the incorporation of contrastive loss. This loss function enhances the model's capacity to differentiate between feature representations by learning to distinguish between similarities and differences between samples.

In our earlier proposed contrastive learning framework, the contrastive loss function enhances the model's ability to capture users' long-term and short-term preferences by systematically comparing positive samples with corresponding negative samples. This innovative approach not only boosts the accuracy of POI recommendation but also substantially enhances the model's robustness across diverse application scenarios and varying user groups. To further improve the generalization capability and the quality of feature representation, the final overall loss function is defined by combining cross-entropy loss with contrastive loss. The formal definition of the total loss function is as follows:

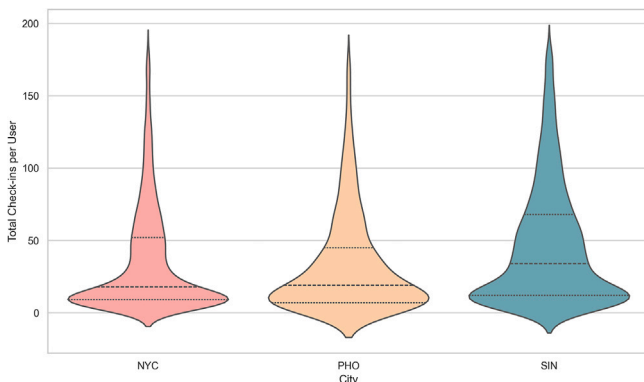
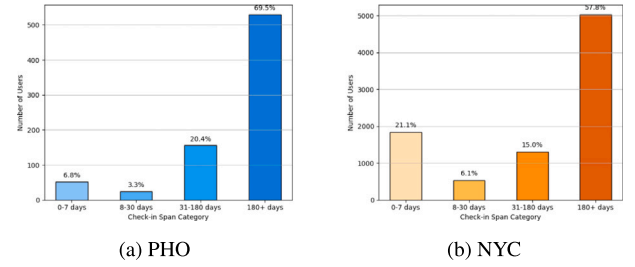
$$\mathcal{L}_{total} = \mathcal{L}_{poi} + \mathcal{L}_{con} \quad (19)$$

5. Experiments

5.1. Datasets and ground truth

To assess the effectiveness of our method, we tested the model on three widely acknowledged real-world benchmark datasets. The datasets, sourced from Foursquare, include data from three cities: Phoenix (PHO), Singapore (SIN), and New York City (NYC). Detailed statistics for these datasets are available in Table 2.

To ensure the reliability of the data, preprocessing is carried out according to the method of [24,45]. Points of Interest (POIs) with fewer than 10 interactions and trajectories containing fewer than 3 check-ins are removed. Inactive users, defined as those with fewer than 5 trajectories, are also excluded to ensure focus on engaged user behavior. The remaining user trajectories are then split into training, validation, and test sets chronologically, using an 8:1:1 ratio to preserve temporal order for accurate sequence modeling. These datasets exhibit diverse user

**Fig. 3.** Distribution of total check-ins per user.**Fig. 4.** User check-in span distribution.

behaviors and realistic long-tail distributions in activity (Fig. 3), providing sufficient historical depth for long-term modeling (Fig. 4) and a challenging testbed for evaluating model generalization under sparse conditions.

5.2. Baselines

To validate the performance of DLSP, we selected three types of models for comparison.

General sequential models.

- **ATST-LSTM [47]**: It utilizes a multi-modal approach to encode POIs along with spatial-temporal contexts, enabling the prediction of the next POI.
- **IMTL [48]**: It utilizes a two-channel encoder mechanism to effectively learn the sequential connections between user activities and corresponding locations.
- **CTLE [49]**: It is developed using a bi-directional Transformer architecture as its foundational framework.
- **GETNext [2]**: It constructs a global POI graph and makes predictions by combining category and time.

Distinguish long- and short-term preference models.

- **PLSPL [6]**: It incorporates two distinct and complementary modules designed to independently capture the user's historical and present preferences.
- **LSPL [11]**: It combines an attention mechanism to capture long-term preferences and two LSTMs for location and category dynamic preferences.
- **CFPRec [45]**: A context-aware recommendation system is proposed to predict future user preferences using a self-ensembling method in dynamic scenarios.

Contrastive learning-based sequential models.

- **ContrRec [50]**: It proposes a contrastive learning task focused on context pairs, ensuring that sequences with identical target items produce comparable outputs.
- **DisenPOI [43]**: It utilizes two distinct graphs to explicitly model sequential and geographical relationships, leveraging self-supervised learning to disentangle their respective influences.
- **CLSPRec [7]**: A novel model for contrastive learning of long-term and short-term preferences that integrates a shared trajectory encoder.

5.3. Metrics

Normalized Discounted Cumulative Gain (NDCG) and Hit Ratio (HR) are key metrics in POI recommender systems. NDCG evaluates ranking accuracy by prioritizing relevant items at the top of the recommendation list, reflecting alignment with user preferences. In contrast, HR focuses on recall, measuring whether the list contains items of interest regardless of order. By combining NDCG and HR, we achieve a comprehensive evaluation, with NDCG reflecting the model's ability to prioritize relevant

Table 3

The performance comparison in HR@K and NDCG@K across three datasets shows that our model achieves the highest results at all thresholds. In each column, the top result is **bolded**.

Model	PHO				SIN				NYC			
	HR@5	HR@10	NDCG@5	NDCG@10	HR@5	HR@10	NDCG@5	NDCG@10	HR@5	HR@10	NDCG@5	NDCG@10
ATST-LSTM [47]	0.1579	0.2377	0.1033	0.1385	0.1296	0.1933	0.1027	0.1476	0.1667	0.2031	0.0912	0.1638
IMTL [48]	0.1830	0.2747	0.1301	0.1632	0.1505	0.1801	0.1051	0.1423	0.1798	0.2422	0.0989	0.1861
CTLE [49]	0.2632	0.3605	0.1995	0.2068	0.2041	0.2784	0.1313	0.1556	0.2421	0.3205	0.1513	0.1841
GETNext [2]	0.5130	0.6147	0.3730	0.4019	0.3517	0.4294	0.2889	0.3163	0.2916	0.3690	0.2119	0.2342
PLSPL [6]	0.1775	0.2569	0.1285	0.1538	0.1447	0.1719	0.1126	0.1384	0.1741	0.2413	0.0961	0.1587
LSPL [11]	0.2303	0.3355	0.1824	0.2493	0.3180	0.3765	0.2398	0.2588	0.2371	0.3122	0.2158	0.2431
CFPRec [45]	0.3421	0.4253	0.2432	0.2730	0.2310	0.3085	0.1588	0.1836	0.2771	0.3606	0.1971	0.2190
ContrRec [50]	0.3381	0.3680	0.2843	0.2939	0.2047	0.2710	0.1454	0.1660	0.1951	0.2368	0.1425	0.1560
DisenPOI [43]	0.4209	0.4988	0.3402	0.3652	0.2808	0.3420	0.2307	0.2506	0.2940	0.3634	0.2401	0.2594
CLSPRec [7]	0.5368	0.6368	0.3811	0.4175	0.3545	0.4093	0.2794	0.2942	0.3545	0.4352	0.2653	0.2870
DLSP(ours)	0.6052	0.6842	0.4511	0.4775	0.4098	0.4826	0.3274	0.3500	0.3740	0.4491	0.2777	0.3051

POIs and HR highlighting its effectiveness in identifying user-preferred locations.

5.4. Settings

We implement the model using the PyTorch framework. Our experiments are run on a 64-bit Linux server equipped with Intel(R) Core(TM) i7-13700F CPUs and NVIDIA RTX 4090 GPUs. For the PHO, SIN, and NYC datasets, the embedding dimension D is uniformly set to 60, the learning rates are all set to $1e-5$, the dropout value of the MLP is set to 0.2, the number of GRU layers is 2 and the number of channels of the TCN is [128,128,64,300].

5.5. Experimental results

Table 3 presents the experimental results comparing the DLSP model with several benchmark models. The results indicate that independently modeling long-term and short-term user preferences significantly outperforms traditional sequence-based approaches. Models leveraging contrastive learning to disentangle and represent distinct user feature dimensions demonstrate superior performance in terms of HR and NDCG across three benchmark datasets.

Our DLSP model combines these advantages through a novel fusion strategy that captures both the dynamic nature of user preferences and the diversity of feature representations. This design leads to notable improvements in recommendation accuracy and sequence quality, achieving an average increase of 10.5 % in HR and NDCG across the three datasets. The key factors contributing to this performance are: (1) Long-term and short-term preferences are modeled as structurally distinct yet complementary. Contrastive learning enhances their representational discrimination without enforcing opposition, enabling more adaptive and robust user preference modeling. (2) A similarity-based fusion mechanism dynamically integrates these preferences, preserving their individual characteristics while achieving complementary information fusion. (3) Incorporating user-specific feature embeddings alongside the fused preference representation enables accurate modeling of past behavioral trajectories. These findings highlight the effectiveness of our contrastive recommendation framework in generating meaningful and actionable representations.

5.6. Ablation experiments

Ablation experiments are conducted to assess the contribution of each component in the DLSP model, focusing on their impact on overall performance. The following variants of the model are tested:

- **DLSP/user**: The model without the user enhancement component.
- **DLSP/cl**: The model without the contrastive learning component.
- **DLSP/fusion**: The model without the fusion component.
- **DLSP**: The full model we propose.

As shown in Fig. 5(a) and (b), the full DLSP model achieves the highest hit rate at HR@5 and HR@10 on both the NYC and SIN datasets. Incorporating both user enhancement and contrastive learning components enables DLSP to effectively capture and integrate user preferences, delivering superior performance. The removal of the user enhancement component (DLSP/user) causes a noticeable drop in accuracy, underscoring the importance of personalized user information in generating precise recommendations. Similarly, omitting the contrastive learning component (DLSP/cl) significantly reduces performance, highlighting its role in disentangling and leveraging shared user behavior patterns. The fusion mechanism (DLSP/fusion) also proves critical; its absence results in substantial performance degradation, demonstrating the necessity of dynamically integrating long-term and short-term preferences. NDCG results in Fig. 5(c) and (d) exhibit a similar trend. The full DLSP model consistently outperforms its ablated versions, achieving superior ranking accuracy across datasets. These findings confirm that each component of DLSP plays a pivotal role in enhancing both the precision and adaptability of POI recommendation.

5.7. Analysis of disentangling preferences

To display the performance of our proposed contrastive learning framework, we adopt the t-SNE (t-Distributed Stochastic Neighbor Embedding) [51] dimensionality reduction technique, which maps the complex data structures in the high-dimensional feature space to the easy-to-visualize two- or three-dimensional space to visually display the learned feature representations. Specifically, for two independent datasets, SIN and NYC, we evaluate the effect of not using contrastive learning, using the CFPRec model and our model. Furthermore, we compare the distinctions in the distributions of long- and short-term preference features among these three models.

Without the use of contrastive learning (Fig. 6(a) and (d)), the distribution of short-term states (blue points) and long-term states (orange points) in the SIN and NYC datasets is very confusing, with little clear distinction between the two. The large overlap between the blue and orange dots, which is particularly noticeable in the New York dataset, suggests that there is a great deal of confusion in the model's treatment of long- and short-term preferences. This situation suggests that, without the use of contrastive learning, the model faces challenges in effectively capturing users' long- and short-term preferences, resulting in a lack of accuracy and personalization in the recommendation results. For the CFPRec (Fig. 6(b) and (e)), although the contrastive loss function is not used, the model processes the user's preferences via two encoders. The results show that the short-term state (blue dots) has a more concentrated cluster distribution compared to the long-term state (orange dots). However, long-term preferences remain more dispersed and are not clearly bounded by short-term preferences. Although the CFPRec model has shown improvements in capturing users' short-term behavior, it is still deficient in handling the integration of long-term

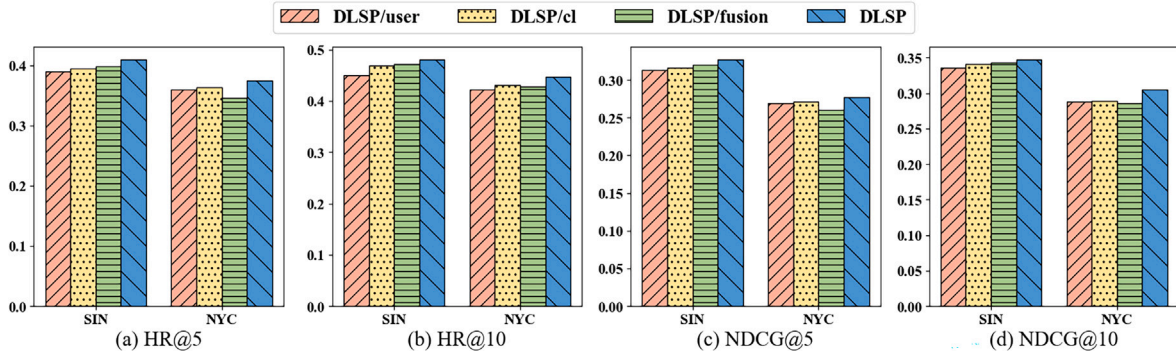


Fig. 5. Ablation experiments of NYC and SIN datasets.

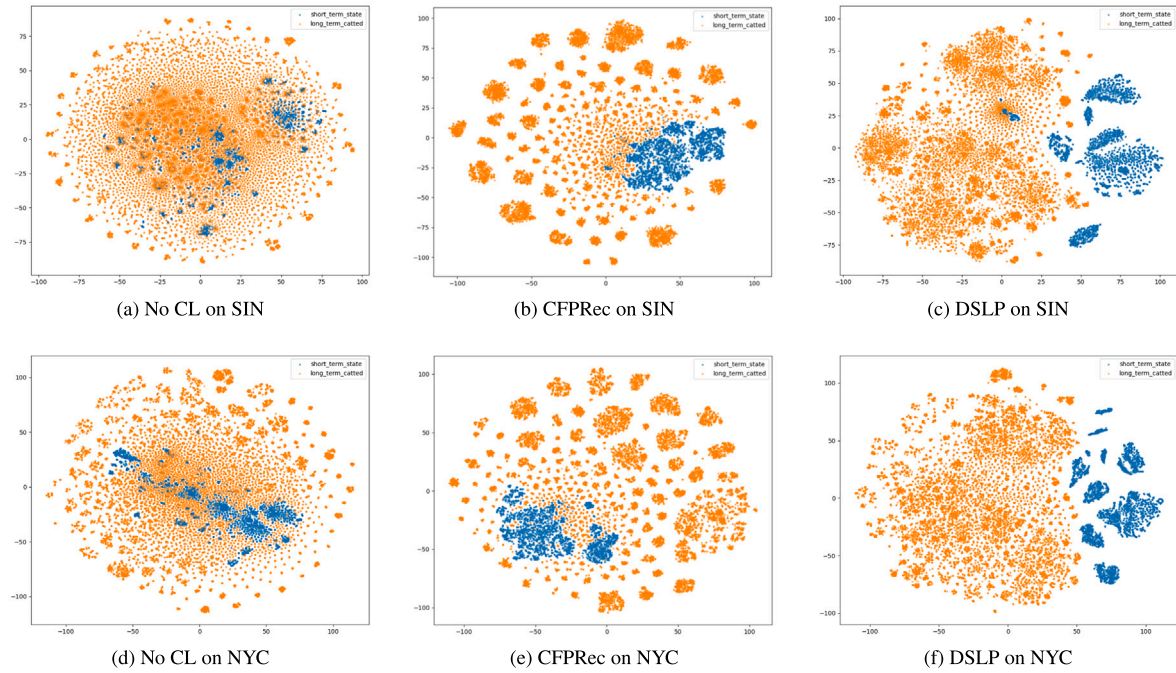


Fig. 6. Comparison of different scenarios. The figure shows the difference in performance between the different models (no contrastive learning, CFPreC, our model) on the SIN and NYC datasets. The orange points represent the long-term state of the user and the blue points represent the short-term state of the user. It can be seen that the different models differ significantly in their ability to separate long-term and short-term states.

preferences with overall user preferences. This limitation arises from the lack of further guidance or optimization on the contrastive loss function, ultimately resulting in restricted capacity to deliver personalized and context-aware recommendations.

Our proposed model (Fig. 6(c) and (f)), on the other hand, significantly improves the separation of short-term and long-term states. In both the NYC and SIN datasets, short-term states (blue dots) and long-term states (orange dots) form clearer clusters, and the boundary between them is very distinct. Contrastive learning significantly enhances the model's capacity to capture dynamic user behavior by explicitly distinguishing between long- and short-term preferences. This enhanced separation effect ensures the accuracy of the recommendation results, which can provide users with more personalized and tailored POI recommendation, further improving the model's generalization performance and recommendation effectiveness.

5.8. Similarity analysis

To verify the effectiveness of the proposed contrastive learning mechanism in long-term and short-term preference modeling, we conduct an in-depth embedding similarity analysis on the PHO dataset. The results

are shown in Fig. 7 and similar effects are also present in the other two datasets. We focus on examining the cosine similarity relationships among four representations: the encoded short-term representation h_s , the encoded long-term representation h_l , the short-term representation p_s obtained by average pooling and the long-term representation p_l obtained by average pooling.

In the basic model without introducing the contrastive learning mechanism, the similarity between the long-term and short-term representations is relatively high. Specifically, $\text{sim}(h_l, h_s) = 0.43$ indicates that there is a significant phenomenon of aliasing long- and short-term preferences in the feature space. In contrast, after the introduction of the contrastive learning mechanism, the similarity distribution has changed. As shown in Fig. 7(b), the similarity decreased to -0.41 , showing a strong negative correlation and improving the degree of decoupling of the characteristics of the encoded long-term and short-term preferences. Meanwhile, both $\text{sim}(h_l, p_l) = 0.18$ and $\text{sim}(h_s, p_s) = 0.26$ have been improved, indicating that the proposed proxy-supervised contrastive learning strategy improves the consistency of the representation between the encoder and its preference. This result is in line with the comparison strategy we designed in Eqs. (9) and (10).

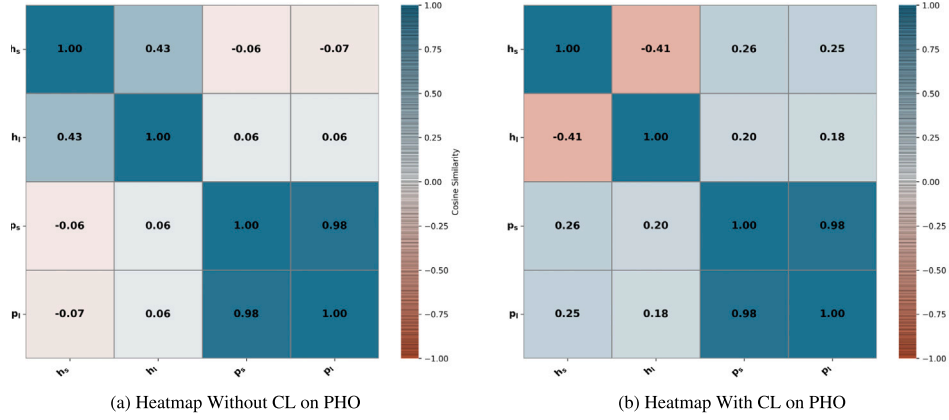


Fig. 7. Heatmap among four representations.

Table 4
Ablation study on different preference fusion mechanisms. The best results are highlighted in bold.

Fusion Mechanism	PHO	
	HR@10	NDCG@10
Dot Product	0.6316	0.4330
MLP	0.6053	0.3938
Attention	0.6579	0.4558
Cosine	0.6842	0.4775

perform hyperparameter tuning to ensure optimal performance across all datasets. Table 4 presents the results of these experiments. It can be observed that the cosine similarity mechanism consistently achieves superior performance across two evaluation metrics (HR@10 and NDCG@10) on the PHO dataset. This demonstrates the effectiveness of the cosine similarity dynamic fusion in balancing predictive accuracy and computational efficiency when fusing long- and short-term user preferences.

5.10. Hyper-parameter sensitivity experiment

We conducted a single-variable sensitivity analysis to isolate the impact of key hyperparameters, with the results shown in Fig. 8. We observe that increasing the embedding dimension generally improves performance up to a certain point, after which gains diminish. For instance, the model achieves its optimal performance on the PHO dataset with an embedding size of 60, indicating that a moderately sized dimension best captures the feature space without overfitting.

5.9. Validation of preference fusion

To empirically validate our proposed preference fusion design, we conduct controlled ablation experiments. Specifically, we replace the proposed cosine similarity dynamic fusion, including Dot Product, a simple MLP and an Attention module. For each variant, we

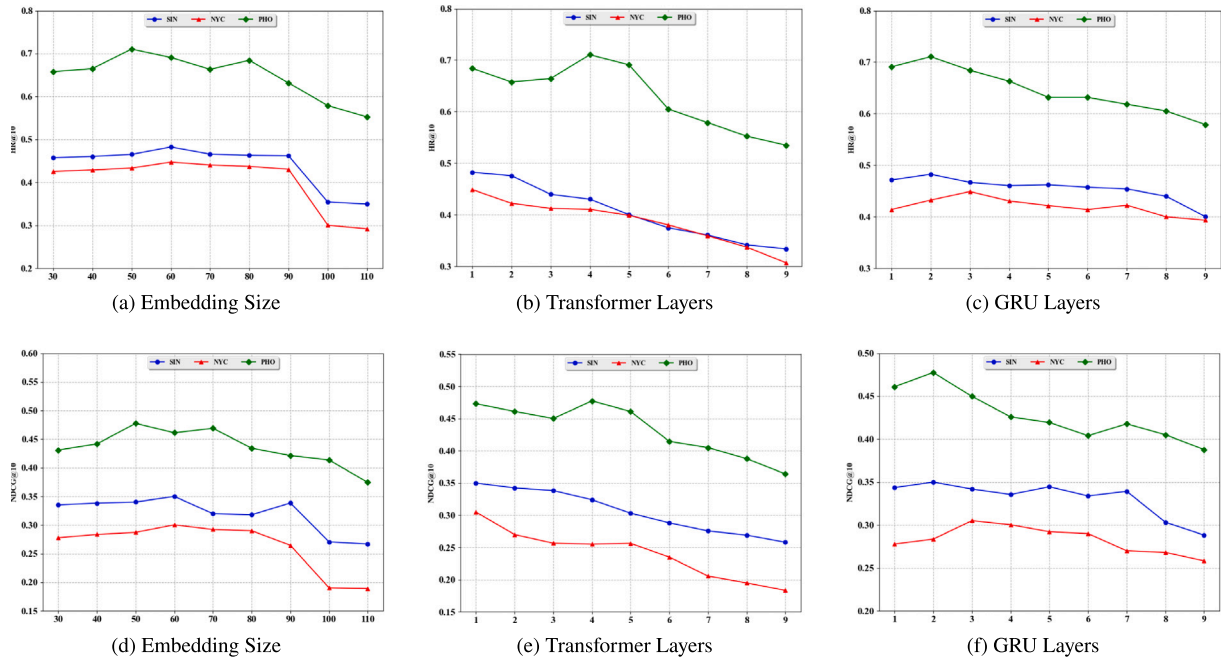


Fig. 8. The effect of hyperparameters on HR@10 (top) and NDCG@10 (bottom) across SIN, NYC, and PHO datasets. (a) Shows performance varying with embedding size. (b) and (c) demonstrate decreasing performance as the number of Transformer and GRU layers increases.

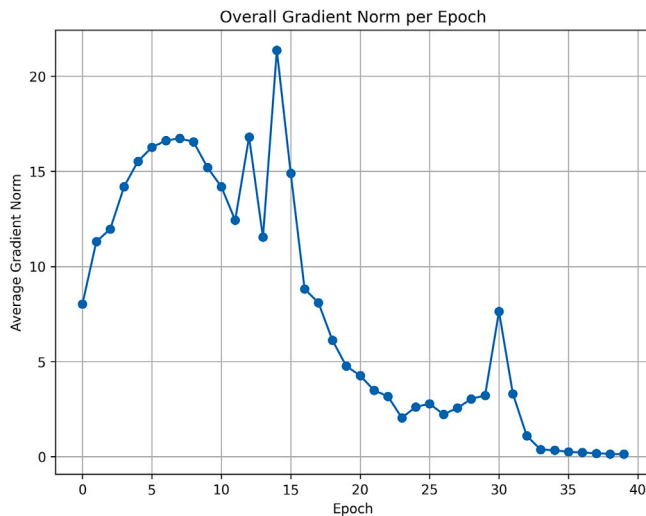


Fig. 9. Training instability of a deep model variant (e.g., 9 layers) on the PHO dataset. The plot shows the average gradient norm per epoch, revealing initial gradient explosion followed by subsequent gradient vanishing.

Conversely, we found that increasing the number of Transformer and GRU layers consistently leads to performance degradation across all datasets. To diagnose the root cause, we trained a deeper variant of our model and plotted its overall gradient norm per epoch. As shown in Fig. 9, the plot reveals severe training instability. The initial epochs are marked by erratic and high gradient norms, a symptom of gradient explosion, while in later epochs, the norm collapses to near-zero, indicating gradient vanishing. This instability prevents the model from converging effectively and is the primary cause of the performance degradation in overly deep architectures. This analysis confirms that a shallower architecture provides a more stable training process and better generalization for this task.

6. Conclusion

This paper presents the DLSP model, a novel approach to capturing and integrating users' long-term and short-term preferences for next POI recommendation. The model employs separate encoders to independently process long-term and short-term preferences, and a contrastive learning framework to disentangle these preferences, revealing their intrinsic connections and distinctions. A similarity-based fusion mechanism dynamically integrates these preferences, achieving a balanced and adaptive representation. Additionally, the user's basic feature information is combined with the fused preferences to construct a comprehensive user model, enabling accurate modeling of past behavioral trajectories and improving prediction performance. Extensive experiments on three benchmark datasets demonstrate the superior effectiveness of DLSP compared to state-of-the-art methods.

Despite its strengths, the experimental results highlight that sequence contrastive learning struggles with datasets exhibiting high sparsity. Future research should focus on developing strategies to address this challenge, potentially by incorporating advanced data augmentation techniques or alternative learning paradigms.

CRedit authorship contribution statement

Jianqun Cui: Supervision, Project administration, Funding acquisition. **Jianing Zhang:** Writing – original draft, Visualization, Validation, Data curation, Conceptualization. **Min Wang:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **Yanan Chang:**

Supervision, Funding acquisition. **Feng Huang:** Writing – original draft, Validation. **Wei Dai:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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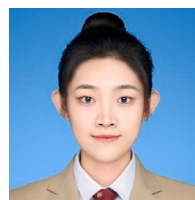
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